

Motion Prediction for Autonomous Driving: Controlled Architectural Ablations of SHARP

Seyed Mohammad Salehi

Master of Science

from the

University of Surrey



School of Computer Science and Electrical and Electronic
Engineering

Faculty of Engineering and Physical Sciences

University of Surrey

Guildford, Surrey, GU2 7XH, UK

September 2026

Supervised by: Prof. Saber Fallah and Kevin Wells

DECLARATION OF ORIGINALITY

I confirm that the project dissertation I am submitting is entirely my own work and that any material used from other sources has been clearly identified and properly acknowledged and referenced. In submitting this final version of my report to the JISC anti-plagiarism software resource, I confirm that my work does not contravene the university regulations on plagiarism as described in the Student Handbook. In so doing I also acknowledge that I may be held to account for any particular instances of uncited work detected by the JISC anti-plagiarism software, or as may be found by the project examiner or project organiser. I also understand that if an allegation of plagiarism is upheld via an Academic Misconduct Hearing, then I may forfeit any credit for this module or a more severe penalty may be agreed.

Motion Prediction for Autonomous Driving: Controlled Architectural Ablations of SHARP

Seyed Mohammad Salehi (URN 6908940)

Author Signature

Date:

Supervisors: Prof. Saber Fallah and Kevin Wells

WORD COUNT

Number of Pages: 55.

Number of Words: 8,194 (abstract and Chapters 1–5; references, appendices, tables and figures excluded).

ABSTRACT

Motion forecasting enables an autonomous vehicle to anticipate several plausible futures for surrounding road users rather than react only to their observed positions. The task is difficult because traffic behaviour is interactive, map-constrained and intrinsically multi-modal, while an operational predictor must update continuously as new observations arrive. This dissertation investigates how the accuracy of SHARP, a short-window streaming motion-forecasting architecture, changes under controlled modifications to its attention, context and temporal encoding components.

The study first establishes local SHARP reference results on Argoverse 2 and then conducts three linked experimental stages. A ten-variant, 20-epoch screening study evaluates gated memory, cross-window consistency, learned temporal pooling, uncertainty-aware context, relative-geometry attention bias, a kinematic stem, endpoint refinement, lane topology and complete temporal-attention replacement by Mamba. Under identical controls, uncertainty-aware target context and relative-geometry bias are the only variants to improve minimum average displacement error over six modes (minADE6), reaching 0.749961 and 0.749679 respectively against 0.752339 for the shortened baseline. A separate 80-epoch attention study compares standard multi-head attention, query-key normalisation (QKNorm), Talking-Heads attention and their combination. QKNorm gives the best minADE6, 0.669777, improving its local control by 0.003683 (0.55%). Mamba placement experiments show that a residual bidirectional module within the chronological agent-history encoder is preferable to scanning pooled scene tokens, although neither tested addition surpasses the longer unmodified SHARP reproduction.

These findings motivate a final confirmatory protocol comprising an article-aligned SHARP baseline, a QKNorm-uncertainty-geometry model, and the same model with a small residual temporal-agent Mamba module. Its result fields are deliberately reserved for validated checkpoints rather than inferred from incomplete training. The dissertation contributes a reproducible ablation framework, a placement-sensitive analysis of Mamba, and evidence that modest geometric and logit-normalisation changes are more reliable than wholesale architectural replacement.

CONTENTS

Declaration of Originality	ii
Word Count	iii
Abstract	iv
List of figures	ix
1 Introduction	1
1.1 Context and motivation	1
1.2 Problem statement	2
1.3 Research aim and questions	3
1.4 Objectives	4
1.5 Contributions	4
1.6 Scope and exclusions	5
1.7 Dissertation structure	6
2 Background and Related Work	7
2.1 Motion-forecasting formulation	7
2.2 Evaluation metrics	7
2.3 Datasets	8
2.3.1 Argoverse 2	8
2.3.2 Argoverse 1 and nuScenes	9
2.4 From recurrent interaction models to vectorised scenes	9
2.5 Transformer attention	10
2.5.1 Query-key normalisation	10
2.5.2 Talking-Heads attention	11
2.5.3 Relative-geometry bias	11
2.6 State-space models and Mamba	11

2.7	Streaming motion forecasting	12
2.8	Research gap	13
3	Methodology and Implementation	14
3.1	Experimental strategy	14
3.2	Reference SHARP architecture	14
3.2.1	Short-window streaming	15
3.2.2	Agent, map and scene encoding	16
3.3	Article-code alignment	16
3.4	Lab 2 reproduction and Mamba placement	17
3.4.1	Scene-token Mamba addition	17
3.4.2	Residual temporal-agent Mamba addition	17
3.4.3	Complete temporal-attention replacement	18
3.5	Lab 2 controlled 20-epoch screen	19
3.5.1	Confidence-gated streamed memory	19
3.5.2	Cross-window consistency	19
3.5.3	Learned temporal pooling	19
3.5.4	Uncertainty-aware target context	20
3.5.5	Relative-geometry attention bias	20
3.5.6	Kinematic motion stem	20
3.5.7	Endpoint refinement	20
3.5.8	Lane topology graph	21
3.6	Lab 3 attention study	21
3.7	Final confirmatory three-run design	21
3.8	Reproducibility, logging and failure recovery	22
3.9	Research ethics and responsible reporting	23
4	Results and Discussion	24
4.1	Reporting convention	24

4.2	SHARP article and Lab 2 reproduction	24
4.3	Mamba placement in SHARP	25
4.4	Lab 2 20-epoch screening study	26
4.5	Lab 3 attention study	28
4.6	Cross-dataset Lab 2 evidence	29
4.7	Training-time context	29
4.8	Final confirmatory study: reserved results	30
4.9	Answers to the research questions	31
4.10	Threats to validity	31
5	Conclusions and Future Work	33
5.1	Summary of the investigation	33
5.2	Contributions	34
5.3	Practical conclusions	34
5.4	Limitations	35
5.5	Future work	36
5.6	Final statement	36
	Bibliography	38
6	Reproducibility and Supporting Evidence	42
6.1	Template and source layout	42
6.2	Full Lab 2 screening metric vector	42
6.3	Mamba implementation parameters	43
6.4	Completed Lab 3 checkpoint evidence	43
6.5	Final-suite validation checklist	43
6.6	Reproduction caveats	44
6.7	Reference and link validation	44
6.8	Suggested final artifact set	45

LIST OF FIGURES

3.1	Experimental programme. Solid boxes are completed evidence; the outlined final stage is reserved for the active confirmatory suite.	14
3.2	Functional overview of SHARP retained by all reported ablations. Architectural changes are confined to labelled local operators; short windows, identity-aware memory, target context and trajectory relay remain present.	15
3.3	Mamba configurations evaluated in SHARP. The residual temporal-agent placement preserves all four original attention blocks; the full replacement removes them.	18
4.1	Best retained minADE6 for unmodified SHARP and the two longer Mamba additions. The baseline differs in schedule; scene-token and temporal-agent Mamba share the 80-epoch, global-batch-32 setting.	25
4.2	Best minADE6 in the Lab 2 20-epoch screen. The first bar is the matched shortened baseline; lower bars are better.	27
4.3	Best minADE6 for the Lab 3 attention operators. QKNorm is the only operator to improve the matched MHA control on this selection metric. . .	28
4.4	Approximate wall-clock hours for completed Lab 2 and Lab 3 reference runs. Differences include model, epoch, batch and system-load effects. . . .	30
4.5	Reserved confirmatory-suite comparison.	31

1 INTRODUCTION

1.1 Context and motivation

An autonomous vehicle does not make decisions from a static picture of the road. It repeatedly receives observations from perception, updates its estimate of the surrounding scene and selects an action while other road users continue to move. Motion forecasting is the component that bridges perception and planning: from an observed history, it estimates where nearby vehicles, cyclists and pedestrians may travel during a future horizon. A planner can then reason about potential conflicts before they occur. Forecasting is therefore not simply a curve-fitting exercise. It must represent interaction, road geometry, uncertainty and timing in a form that can be queried repeatedly under real-time constraints.

The future of a road user is not unique. A vehicle approaching a junction may turn, proceed or yield; a pedestrian may cross or wait. A useful predictor consequently returns several trajectories and an associated confidence for each one. This property is referred to as multimodality: one observed history corresponds to multiple physically and socially plausible futures. Forecasting must also account for interactions. A target vehicle’s future depends on relative position, heading, right-of-way and the likely actions of neighbouring agents. High-definition map elements constrain these possibilities further. The combined problem is difficult because a model must distinguish plausible alternatives without averaging them into an implausible path.

Most public benchmarks formulate motion forecasting as independent snapshots with a fixed observation duration. This formulation is valuable for standardised comparison, but it differs from a deployed system in which predictions are refreshed continuously. Treating each refresh as an unrelated scene duplicates computation and discards useful historical state. RealMotion addressed this mismatch by carrying context and prior trajectories across consecutive scenes [16]. SEAM subsequently introduced endpoint-aware streaming [11]. SHARP further focused on agents with heterogeneous observation lengths: it processes non-overlapping short windows and transfers latent context between matched instances [10]. These methods shift the research question from “How accurately can a

fixed history be encoded?” to “How can useful state be accumulated without weakening accuracy when observations arrive incrementally?”

SHARP is an appropriate basis for this dissertation because its central contribution is both practically relevant and sufficiently compact for controlled architectural investigation. The published model contains approximately 4.6 million parameters and reports competitive Argoverse 2 performance with low inference latency [10]. More importantly, its streaming innovation is localised: short observation windows, instance-aware context transfer, target-centric context and trajectory relay are identifiable mechanisms that can be preserved while attention or temporal encoding is modified. This makes it possible to ask whether a proposed improvement helps forecasting accuracy rather than merely replacing the problem formulation.

1.2 Problem statement

Recent sequence models offer several plausible ways to improve SHARP. Transformer attention can represent direct pairwise interaction, but its logits may become excessively sharp when query and key magnitudes grow. Query–key normalisation (QKNorm) removes that magnitude dependence and learns the logit scale explicitly [7]. Relative-geometry bias can expose displacement, range and heading differences directly to scene attention rather than requiring them to be reconstructed from token content. Prediction uncertainty can also inform how widely a later streaming window searches around each previously predicted endpoint.

Mamba provides another alternative. It is a selective state-space model (SSM) whose recurrent parameters depend on the input, allowing information to be retained or forgotten while scaling linearly with sequence length [5]. DeMo demonstrated that Mamba can be useful in motion forecasting when it is assigned to semantically ordered agent histories and future-state sequences [19]. This does not imply that inserting Mamba anywhere in SHARP will improve it. A state-space scan assumes an order with meaningful recurrence. The ordering of pooled agent and lane tokens is not equivalent to time, and SHARP’s ten-step local history is much shorter than the sequences for which Mamba is normally motivated. Placement and residual strength therefore require empirical study.

The practical problem addressed here is to determine which local architectural changes improve SHARP under controlled training, while preserving its defining short-window streaming behaviour. A second problem is methodological: long multi-GPU experiments are vulnerable to interruption, distributed-training faults and subtle configuration drift. Conclusions are only credible if each comparison records the actual global batch, schedule, checkpoint-selection rule and evaluation split, and if incomplete runs are not mistaken for final results.

1.3 Research aim and questions

The aim of this dissertation is to evaluate targeted attention, context and state-space modifications to SHARP for multimodal motion forecasting, using reproducible controlled ablations on Argoverse 2 while retaining the original streaming architecture.

The investigation is organised around four research questions:

1. RQ1: How closely can the released SHARP model be reproduced on Argoverse 2, and which differences between the article, public configuration and local distributed environment affect that comparison?
2. RQ2: Which of a broad set of local architecture modifications show evidence of improved displacement accuracy under an identical shortened training schedule?
3. RQ3: Does changing the attention operator from standard multi-head attention to QKNorm, Talking-Heads attention, or their combination improve SHARP under a full 80-epoch controlled study?
4. RQ4: Does Mamba benefit SHARP more as a residual chronological agent-history module than as a scene-token addition or a complete replacement for temporal attention?

The final three-run study described in Chapters 3 and 4 is a confirmatory extension of these questions. It tests whether individually promising changes remain beneficial when combined at the full article schedule. Its metric fields are reserved until validated output is available.

1.4 Objectives

The work was divided into the following objectives:

1. establish an evidence-backed description of the SHARP article and released implementation, including data representation, streaming state, optimisation and evaluation;
2. reproduce SHARP on Argoverse 2 and retain comparable validation metrics, logs, checkpoints and environment manifests;
3. implement a controlled 20-epoch screen of eight proposed accuracy modifications and one complete temporal Mamba replacement, each isolated from the others;
4. implement an 80-epoch attention study in which only the attention operator changes;
5. compare two residual Mamba placements and one complete temporal-attention replacement without removing SHARP’s instance-aware context stream;
6. design a final full-length comparison combining only changes supported by the preceding evidence;
7. make long-running experiments recoverable through per-epoch checkpoints, bounded preflight tests, warning scans and compact GitHub artifacts; and
8. report negative as well as positive results, distinguishing a screening result, a best checkpoint, a final-epoch metric and an article value.

1.5 Contributions

The principal contribution is not a claim that a single new layer universally improves motion prediction. It is a controlled analysis of where modest changes help or harm a compact streaming predictor. The dissertation makes five specific contributions.

First, it documents the article–code alignment required to train an article-schedule SHARP baseline. The released Hydra configuration and the article supplement do not use identical epoch and learning-rate values. The final protocol pins the released architecture but explicitly applies the article’s 80-epoch schedule, rather than describing the repository defaults as the article setup.

Second, it provides a ten-variant screening study under identical data, seed, global batch and optimisation controls. The result narrows a broad design space: uncertainty-aware target context and relative-geometry bias improve minADE6 slightly, whereas several intuitively attractive additions do not.

Third, it provides a full-length controlled attention comparison. QKNorm improves the best minADE6 over the local MHA control, while Talking-Heads attention and the combined operator do not. This prevents the final design from being selected by architectural novelty alone.

Fourth, it isolates Mamba placement. Scene-token Mamba is applied after chronological detail has already been pooled; residual temporal-agent Mamba is applied within each ordered history; full replacement removes temporal attention entirely. Their outcomes support a restrained conclusion: the chronological residual placement is the strongest Mamba formulation tested, but the unmodified SHARP reproduction remains more accurate.

Fifth, it develops a reproducibility and recovery framework suitable for multi-day training. The framework records exact run directories, emits a last checkpoint every epoch, resumes incomplete variants, evaluates completed checkpoints separately, scans warnings and publishes small textual and graphical artifacts while retaining large checkpoints locally.

1.6 Scope and exclusions

The quantitative dissertation scope is SHARP on Argoverse 2, with secondary Lab 2 validation evidence on Argoverse 1 and nuScenes used only to discuss generalisation and reproduction gaps. Results from the earlier single-A4000 laboratory stage are excluded, as required by the project scope. SEAM is discussed as related work because endpoint-aware streaming helps locate SHARP within the literature; no SEAM experiment is reported. DeMo is used to explain why Mamba is relevant to motion forecasting, but the Mamba modifications developed here remain SHARP ablations rather than DeMo reproductions.

The dissertation does not claim closed-loop vehicle safety, planning improvement or deployment in CARLA/ROS2. These were considered in the initial project planning, but

the completed technical evidence is a controlled forecasting study. Open-loop benchmark metrics are useful but cannot establish collision avoidance, calibration under distribution shift, or downstream planning value by themselves.

1.7 Dissertation structure

Chapter 2 formulates multimodal forecasting, defines the metrics and reviews recurrent, vectorised, attention-based and streaming methods. Chapter 3 details SHARP, the datasets, controlled training protocol, nine screening modifications, attention variants and Mamba placements. Chapter 4 presents the completed Lab 2 and Lab 3 results, analyses cross-study consistency, and reserves the final confirmatory cells. Chapter 5 answers the research questions, states limitations and identifies the next experiments that can be justified by the evidence. The appendix records configurations, provenance and the result-insertion protocol.

2 BACKGROUND AND RELATED WORK

2.1 Motion-forecasting formulation

Consider a traffic scene observed at discrete times $t = -T_{\text{obs}} + 1, \dots, 0$. For agent i , the observation

$$\mathbf{X}_i = \{\mathbf{x}_i^t\}_{t=-T_{\text{obs}}+1}^0 \quad (2.1)$$

contains positions and derived motion features in a local coordinate frame. The scene also contains a vector map $\mathcal{M}$ and observations of neighbouring agents. The forecasting model estimates K future trajectories

$$\hat{\mathbf{Y}}_{i,k} = \{\hat{\mathbf{y}}_{i,k}^t\}_{t=1}^{T_{\text{pred}}}, \quad k \in \{1, \dots, K\}, \quad (2.2)$$

with mode probabilities $p_{i,k}$ satisfying $\sum_k p_{i,k} = 1$. A mode is one distinct future hypothesis. For Argoverse 2, the experiments use $K = 6$ modes and forecast six seconds from two seconds of available history, while SHARP internally divides the observations into shorter streaming windows [18, 10].

The model must resolve three coupled sources of difficulty. Aleatoric uncertainty is uncertainty inherent in the future: more than one manoeuvre can be correct. Interaction means that one agent’s trajectory depends on others. Map conditioning constrains motion through lane direction, boundaries and topology. A deterministic mean trajectory can lie between two valid modes and correspond to neither one. Multimodal forecasting avoids this by predicting a small distribution of trajectories, but evaluation then has to consider both geometric error and confidence.

2.2 Evaluation metrics

Let $\mathbf{Y}_i$ be the ground-truth future and let $\|\cdot\|_2$ denote Euclidean distance. The minimum average displacement error over K modes is

$$\text{minADE}_K = \frac{1}{N} \sum_{i=1}^N \min_k \left(\frac{1}{T_{\text{pred}}} \sum_{t=1}^{T_{\text{pred}}} \|\hat{\mathbf{y}}_{i,k}^t - \mathbf{y}_i^t\|_2 \right). \quad (2.3)$$

It measures the average geometric error of the best available mode. The minimum final displacement error is

$$\text{minFDE}_K = \frac{1}{N} \sum_{i=1}^N \min_k \left\| \hat{\mathbf{y}}_{i,k}^{T_{\text{pred}}} - \mathbf{y}_i^{T_{\text{pred}}} \right\|_2. \quad (2.4)$$

The miss rate (MR) is the proportion of samples for which every mode’s endpoint error exceeds the benchmark threshold τ :

$$\text{MR}_K = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left[\min_k \left\| \hat{\mathbf{y}}_{i,k}^{T_{\text{pred}}} - \mathbf{y}_i^{T_{\text{pred}}} \right\|_2 > \tau \right]. \quad (2.5)$$

For the Argoverse 2 motion-forecasting benchmark, $\tau = 2$ metres [18]. The Brier-adjusted endpoint score adds a confidence penalty:

$$\text{b-minFDE}_K = \frac{1}{N} \sum_{i=1}^N \min_k \left[\left\| \hat{\mathbf{y}}_{i,k}^{T_{\text{pred}}} - \mathbf{y}_i^{T_{\text{pred}}} \right\|_2 + (1 - p_{i,k})^2 \right]. \quad (2.6)$$

All metrics in this dissertation are lower-is-better. Minimum displacement metrics are oracle metrics: they select the best trajectory after observing the ground truth. They therefore measure whether a useful hypothesis is present, not whether the model assigned it the highest confidence. Reporting the Brier-adjusted value and MR alongside minADE and minFDE gives a less one-dimensional view.

Two reporting distinctions are essential. First, validation and test splits are not interchangeable. Second, the best minADE6 checkpoint need not share the other metric values shown at the final epoch. Where a checkpoint filename supplies only minADE6, this dissertation states that limitation rather than combining unrelated values silently.

2.3 Datasets

2.3.1 Argoverse 2

Argoverse 2 (AV2) contains motion-forecasting scenarios with tracked road users and detailed vector maps [18]. It is the primary dataset because it provides enough scale and interaction diversity to test multi-agent context, and because the SHARP article reports both validation and challenge-test results on it. The local pipeline uses SHARP’s processed single-agent train and validation scenarios. Inputs include agent positions, velocities, se-

mantic types and lane polylines. The final architecture embeds these heterogeneous values into a shared 128-dimensional representation.

The evaluation crop and sampling follow the retained SHARP setup. For AV2, each SHARP streaming sub-window contains ten observations, representing one second, and the full local sample provides two such windows. The future contains 60 points over six seconds. Map and agent tokens outside the region of interest are excluded before scene attention. No augmentation or cross-dataset pre-training is used in the controlled studies.

2.3.2 Argoverse 1 and nuScenes

Argoverse 1 introduced a large urban forecasting benchmark with vector maps [3]. nuScenes provides multimodal autonomous-driving data across multiple sensors and locations [2]. Secondary Lab 2 SHARP evaluations on these datasets are included to document reproduction behaviour, not to combine incomparable leaderboard settings. In particular, the retained local nuScenes vector uses six modes while the article challenge row uses five; the split and mode count are stated with every comparison.

2.4 From recurrent interaction models to vectorised scenes

Early learned trajectory predictors relied on recurrent neural networks. Social LSTM assigned a recurrent state to each person and pooled neighbouring hidden states to model interaction [1]. Social GAN added adversarial training and a variety loss to encourage multiple socially plausible trajectories [6]. Trajectron++ combined a graph-structured recurrent model, heterogeneous agent types, map context and optional dynamic constraints [13]. These models established two principles that remain relevant: an agent’s history should be encoded as an ordered sequence, and interaction cannot be reduced to independent per-agent extrapolation.

Rasterising a scene into a bird’s-eye-view image makes convolutional processing convenient but can discard the exact structure of map polylines and add substantial computation. VectorNet instead represents trajectories and map elements as vectors, applies local subgraph encoding and then models global interaction [4]. LaneGCN builds explicit lane graphs and passes messages through actor-to-lane, lane-to-lane, lane-to-actor and actor-to-actor relations [8]. These approaches motivated the vector-token representation used by many later forecasting models, including SHARP.

HiVT decomposes vectorised forecasting into local context extraction and global interaction, while exploiting translation and rotation invariance [20]. QCNet develops a query-centric coordinate system that makes scene representations reusable as the observation window advances [21]. These designs demonstrate that accuracy and online efficiency are affected not only by the decoder but by the reference frame, token hierarchy and re-use of scene computation.

2.5 Transformer attention

For a token matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, scaled dot-product attention first forms queries, keys and values:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V. \quad (2.7)$$

The output is

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}} + \mathbf{B}\right) \mathbf{V}, \quad (2.8)$$

where d_h is the per-head dimension and $\mathbf{B}$ contains masks or learned biases. Multi-head attention repeats this calculation in separate learned subspaces and concatenates the results [17]. A head can specialise in a particular interaction pattern, while the feed-forward and residual layers refine the aggregated representation.

Attention is well suited to motion forecasting because any relevant agent or lane token can interact directly with another. Its weakness is that the dot-product logit depends on both direction and magnitude. Large query or key norms can make the softmax distribution sharply concentrated before that concentration has acquired a useful semantic meaning.

2.5.1 Query-key normalisation

QKNorm normalises each query and key vector and learns a scale g_h for head h [7]:

$$\hat{\mathbf{q}} = \frac{\mathbf{q}}{\max(\|\mathbf{q}\|_2, \epsilon)}, \quad \hat{\mathbf{k}} = \frac{\mathbf{k}}{\max(\|\mathbf{k}\|_2, \epsilon)}, \quad (2.9)$$

$$\ell_h(\mathbf{q}, \mathbf{k}) = g_h \hat{\mathbf{q}}^\top \hat{\mathbf{k}}. \quad (2.10)$$

The logit now depends on angular similarity and a trainable head-level temperature. The hypothesis in this project is not that QKNorm adds scene knowledge, but that it improves optimisation of the attention already responsible for temporal and spatial interaction.

2.5.2 Talking-Heads attention

Standard heads are independent until their outputs are concatenated. Talking-Heads attention introduces learned linear mixing across heads before and after softmax [15]. If $\mathbf{L}$ is the stack of head logits, a simplified representation is

$$\mathbf{A} = \mathbf{P}_{\text{post}} \text{softmax}(\mathbf{P}_{\text{pre}} \mathbf{L}), \quad (2.11)$$

where $\mathbf{P}_{\text{pre}}$ and $\mathbf{P}_{\text{post}}$ mix the head dimension. This allows complementary heads to exchange evidence. The additional flexibility can also be redundant in a compact model, so it requires a controlled experiment rather than an assumption of improvement.

2.5.3 Relative-geometry bias

Transformers require positional information because attention itself is permutation invariant. Relative-position attention shows that pairwise relation features can be incorporated directly into the score [14]. Motion interaction is explicitly geometric, so the Lab 2 bias uses the normalised displacement $\Delta x, \Delta y$, distance r , and periodic heading difference $\cos \Delta \theta, \sin \Delta \theta$. A small multilayer perceptron maps this relation vector to one additive bias per attention head:

$$b_{ij}^h = f_h \left(\frac{\Delta x_{ij}}{30}, \frac{\Delta y_{ij}}{30}, \frac{\|\Delta \mathbf{x}_{ij}\|_2}{30}, \cos \Delta \theta_{ij}, \sin \Delta \theta_{ij} \right). \quad (2.12)$$

The final layer is zero-initialised, so the modified model begins with the baseline attention scores and learns geometry only when it lowers the training objective.

2.6 State-space models and Mamba

A continuous linear state-space model represents a latent state $\mathbf{h}(t)$ driven by input $\mathbf{x}(t)$:

$$\frac{d\mathbf{h}(t)}{dt} = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t), \quad \mathbf{y}(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}\mathbf{x}(t). \quad (2.13)$$

After discretisation, the recurrent form becomes

$$\mathbf{h}_t = \bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}_t \mathbf{h}_t + \mathbf{D} \mathbf{x}_t. \quad (2.14)$$

Mamba makes selected SSM parameters input-dependent, so the recurrence can retain useful information and suppress irrelevant information on the basis of the current token [5]. A hardware-aware selective scan evaluates the recurrence efficiently. Unlike pairwise attention, the computational cost scales linearly with sequence length.

Linear complexity alone is not sufficient reason to use Mamba in SHARP. The observed agent sequence has only ten points per short window, so quadratic attention is not a bottleneck there. The stronger motivation is inductive bias: recurrence matches chronological order and can summarise motion state. Conversely, applying Mamba to an arbitrary list of pooled scene tokens gives the state transition no stable temporal meaning. The project therefore distinguishes three cases: scene-token addition, residual temporal-agent addition and complete temporal-attention replacement.

DeMo is important supporting evidence because it uses unidirectional Mamba for observed histories and bidirectional Mamba in future-state and coupling modules, together with attention [19]. It demonstrates that attention and selective state-space processing can be complementary in motion forecasting. It does not establish that SHARP benefits from the same configuration, because the streaming mechanism, decoder and sequence lengths differ.

2.7 Streaming motion forecasting

Snapshot prediction discards the relationship between consecutive invocations. RealMotion instead accumulates scene context and relays previous trajectories as driving continues [16]. SEAM uses endpoint-aware modelling to retain useful context efficiently [11]. SHARP processes non-overlapping short windows and associates latent state with agent identity, allowing an agent observed briefly in the current window to benefit from earlier matched observations [10].

SHARP’s innovation has three consequences for this dissertation. First, a valid ablation must leave the identity-based context stream intact unless the stream itself is the tested

variable. Second, evaluation should include short and heterogeneous observation histories rather than only one fixed context. Third, a modification that improves a snapshot metric but destroys incremental state reuse is not an uncomplicated improvement. For this reason, the experiments change local operators while preserving windowing, memory transfer, target-centric context, trajectory relay and dual training.

2.8 Research gap

The literature supplies strong individual mechanisms but limited evidence about their interaction inside a compact streaming model. QKNorm has a clear optimisation rationale, relative geometry has a clear scene prior, and Mamba has a clear sequential rationale. What is missing is a controlled placement study that holds SHARP’s streaming system and optimisation fixed. The experimental programme therefore progresses from broad independent screening, through a full-length attention ablation, to a narrow confirmatory combination. This staged design reduces the risk of attributing an improvement to a component that merely benefited from a different schedule, batch or checkpoint rule.

3 METHODOLOGY AND IMPLEMENTATION

3.1 Experimental strategy

The project uses staged controlled ablation rather than introducing all modifications at once. This decision is central to the validity of the conclusions. If attention, context selection, temporal encoding and optimisation changed together, an improved metric could not be assigned to a particular mechanism. The programme shown in Figure 3.1 therefore proceeds from reproduction to broad screening, then to a full-length attention study, then to placement-specific Mamba experiments and finally to a narrow confirmatory combination.

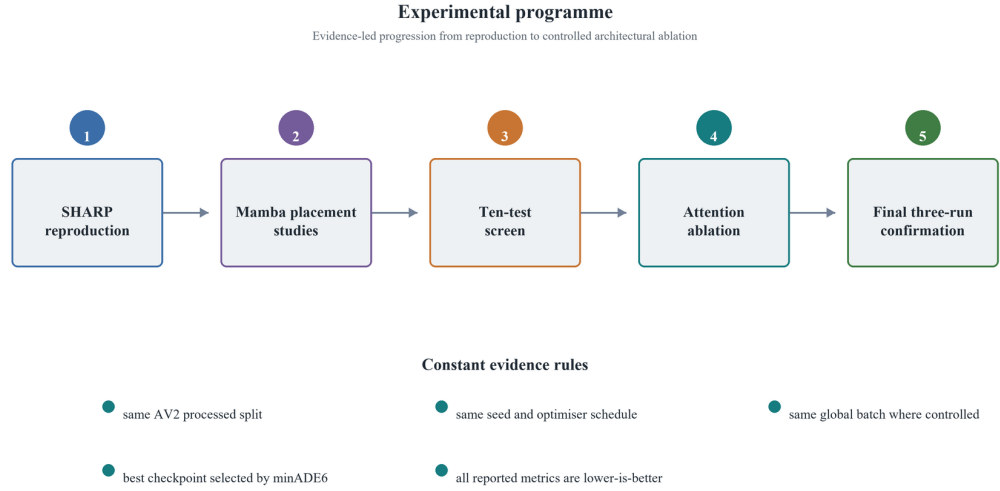


Figure 3.1: Experimental programme. Solid boxes are completed evidence; the outlined final stage is reserved for the active confirmatory suite.

Each scientific run has its own immutable code copy and result directory. A run is considered comparable only when its dataset split, seed, schedule, global batch and metric-selection rule are recorded. Terminal snapshots, recovery attempts and GitHub publication retries are operational evidence, not additional model variants.

3.2 Reference SHARP architecture

The baseline is the single-agent SHARP architecture described by Prutsch et al. [10]. The final-suite source is pinned to audited commit `f6bf2fc0109f9838cdc24bfb763b5c3e6847c2ae`

from the authors’ released repository. The local package records hashes for reviewed source files before applying any compatibility patch [12].

3.2.1 Short-window streaming

Instead of re-encoding a complete two-second history at every invocation, SHARP divides the observation into non-overlapping one-second windows. Each window contains ten points in the AV2 configuration. Let $\mathbf{z}_{i,w}$ be the encoded token for instance i in window w , and let $\mathbf{m}_{i,w-1}$ be the stored context for the same instance from the previous window. When identity i is matched, the current token is updated using the prior latent context. Unmatched agents are processed from their current short history alone. The updated context is stored for the next invocation.

This instance-aware stream differs from carrying one undifferentiated scene state. It preserves a separate history for each tracked road user, which is important when agents enter or leave the field of view. Previous mode predictions also define endpoint-centric target context for the next window, and the trajectory relay supplies previous forecasts to later inference. Dual training applies losses to both streaming and non-streaming paths so that a shared network remains robust to different observation lengths.

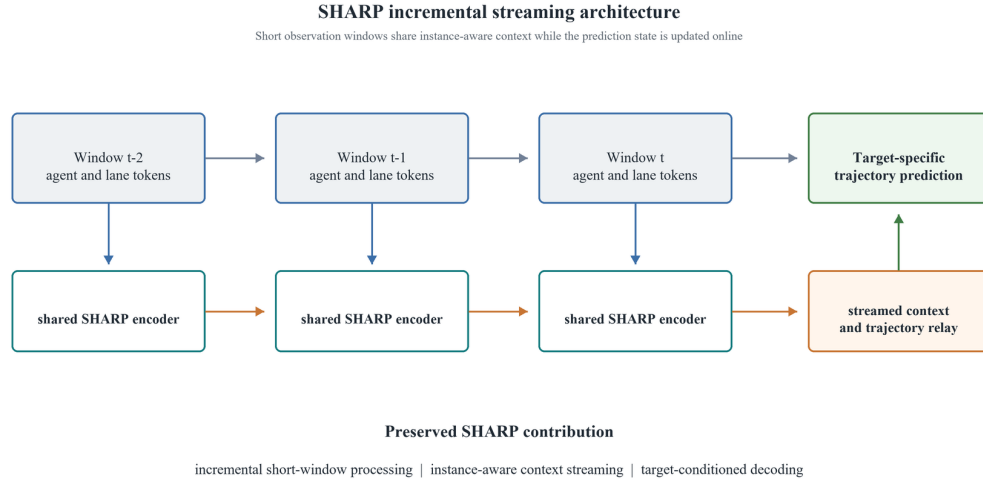


Figure 3.2: Functional overview of SHARP retained by all reported ablations. Architectural changes are confined to labelled local operators; short windows, identity-aware memory, target context and trajectory relay remain present.

3.2.2 Agent, map and scene encoding

Agent positions, velocities and categorical types are projected into a feature dimension $D = 128$. Each agent’s short chronological history passes through four temporal self-attention blocks and masked temporal max pooling. Lane polylines are embedded separately. The pooled agent tokens and lane tokens then pass through four scene-attention blocks, allowing interaction among road users and map elements. All attention uses eight heads, pre-normalisation, GELU feed-forward layers and residual connections. The baseline dropout is 0.2 in the released model.

Target-centric context is constructed around the prior predicted endpoints. Two target-context attention blocks refine each of six mode queries using the scene tokens selected around the relevant endpoint. A multimodal decoder predicts six trajectories and their confidence logits. The training objective combines winner-takes-all Smooth L1 trajectory regression, confidence classification and auxiliary regression for non-target agents, as specified by SHARP [10].

3.3 Article-code alignment

The article supplement specifies one Quadro RTX 8000, global batch 32, 80 AV2 epochs, 13 warm-up epochs, AdamW, a learning rate rising to 10^{-4} then cosine-decaying to 10^{-5} , weight decay, gradient clipping, no augmentation and no cross-dataset training [10]. The audited public Hydra configuration had drifted to 60 epochs and a peak learning rate of 10^{-3} . The final protocol consequently preserves the pinned architecture and data settings but explicitly replaces those two configuration values with the article schedule. Describing unmodified repository defaults as the article setup would be inaccurate.

Table 3.1 lists the controls used for the prepared full-length three-run comparison. Four RTX 2080 Ti GPUs execute distributed data parallel (DDP) training with eight scenarios per rank, preserving global batch 32 without learning-rate scaling. This is a hardware adaptation, not an exact execution reproduction: sample ordering, collective reduction, CUDA kernels and dependency versions can still alter the optimisation trajectory.

Table 3.1: Full-length final-suite training controls.

Setting	Value
Dataset	Argoverse 2, SHARP processed train/validation
Epochs / warm-up	80 / 13
Optimizer	AdamW
Peak / minimum learning rate	10^{-4} / 10^{-5}
Weight decay	10^{-2}
Gradient clipping	Global norm 5
Global batch	32 (8 per rank $\times$ 4 ranks)
Precision	FP32
Seed	2333
Prediction modes	6
Feature dimension / heads	128 / 8
Augmentation / cross-dataset training	None / none
Checkpoint selection	Lowest validation minADE6

3.4 Lab 2 reproduction and Mamba placement

The first local SHARP baseline used 60 epochs, global batch 16 and four RTX 2080 Ti GPUs. It is retained as a reproduction result but is not described as the article’s 80-epoch schedule. Two later 80-epoch Mamba experiments use global batch 32 and otherwise retain the SHARP optimisation controls.

3.4.1 Scene-token Mamba addition

The first Mamba experiment inserts a fused Mamba layer after the per-agent temporal histories have been encoded and pooled. It therefore scans the combined sequence of agent and lane scene tokens. This placement preserves the SHARP temporal attention stack and the streaming context, but the scan order is a data-structure order rather than a physical time axis. Its hypothesis was that selective recurrence might provide a compact global context refinement. Its principal limitation is equally clear: adjacent tokens need not be temporally or topologically adjacent.

3.4.2 Residual temporal-agent Mamba addition

The second placement removes the scene-token Mamba completely and inserts a small bidirectional residual Mamba module between temporal-agent attention blocks 2 and 3. Figure 3.3 contrasts the placements. The temporal-agent sequence is semantically well ordered, and the operation occurs before temporal pooling discards timestep detail.

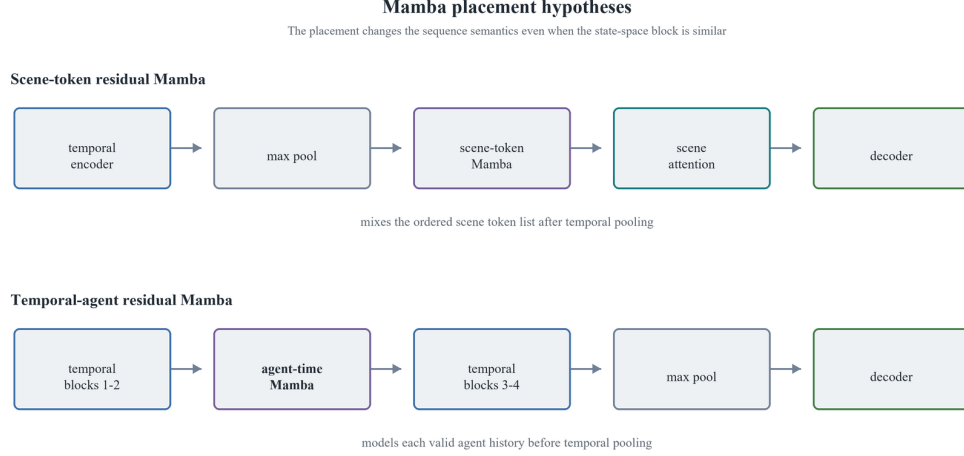


Figure 3.3: Mamba configurations evaluated in SHARP. The residual temporal-agent placement preserves all four original attention blocks; the full replacement removes them.

For valid observations $\mathbf{X}$, separate Mamba modules process the sequence forward and backward. Their outputs are combined with a learned per-channel gate $\mathbf{g} = \sigma(\boldsymbol{\alpha})$:

$$\Delta \mathbf{X} = \mathbf{g} \odot \text{Mamba}_{\rightarrow}(\text{LN}(\mathbf{X})) + (1 - \mathbf{g}) \odot \text{Reverse}[\text{Mamba}_{\leftarrow}(\text{Reverse}[\text{LN}(\mathbf{X})])], \quad (3.1)$$

$$\mathbf{X}' = \mathbf{X} + \gamma \odot \text{Dropout}(\Delta \mathbf{X}), \quad (3.2)$$

where γ is initialised to 0.01. The small LayerScale makes the initial network close to SHARP and allows Mamba to enter gradually. The exact configuration is $d_{\text{state}} = 8$, convolution width 3, expansion 1, dropout 0.1 and 128-dimensional input. Valid timesteps are compacted before recurrence; padded entries are restored afterwards. Agents are processed in chunks of 128 for memory stability, but the time axis is never chunked. Fused CUDA selective scan is used.

3.4.3 Complete temporal-attention replacement

The replacement variant removes all four agent-history attention blocks and uses four unidirectional Mamba blocks with $d_{\text{state}} = 16$, convolution width 4 and expansion 2, followed by the original temporal max pooling. This is a deliberately stronger intervention. It tests whether recurrence can replace temporal attention rather than complement it. Scene attention, context streaming and the decoder remain SHARP components.

3.5 Lab 2 controlled 20-epoch screen

The broad screen uses 20 rather than 80 epochs to compare ten models within available compute time. Every row uses AV2, seed 2333, global batch 32 across four GPUs, AdamW, 13 warm-up epochs, the same learning-rate bounds, SyncBatchNorm and identical checkpointing. Keeping 13 warm-up epochs makes this a deliberately conservative short schedule: only seven epochs remain after peak learning rate. The absolute values must not be presented as full convergence, but the matched design supports within-suite ranking.

Test 1 is unmodified SHARP. Tests 2–9 each introduce only the named change. Test 10 is the complete Mamba replacement described above.

3.5.1 Confidence-gated streamed memory

A learned gate receives the current token, the memory-updated token and their difference. It interpolates between current and updated representations:

$$\mathbf{z}' = \mathbf{z} + \sigma(f([\mathbf{z}, \tilde{\mathbf{z}}, \tilde{\mathbf{z}} - \mathbf{z}])) \odot (\tilde{\mathbf{z}} - \mathbf{z}). \quad (3.3)$$

The intended effect is to suppress stale memory while preserving a reliable historical update.

3.5.2 Cross-window consistency

This variant adds a Smooth L1 penalty, weighted by 0.05, between the current confidence-weighted prediction and the transformed previous prediction. The term encourages adjacent streaming windows to describe a consistent future. It can become harmful if applied before the modes have stabilised, because two uncertain predictions need not align mode by mode.

3.5.3 Learned temporal pooling

Masked attention pooling replaces per-feature temporal maximum. A learned query assigns one weight to each valid timestep, allowing the encoder to prefer braking, turning or recent observations. The trade-off is that hard max pooling can retain a strong feature regardless of its timestep, while one shared temporal weighting may smooth useful extremes.

3.5.4 Uncertainty-aware target context

Let p_k be the previous probability of mode k and let

$$H(\mathbf{p}) = -\frac{\sum_k p_k \log(p_k + \epsilon)}{\log K} \quad (3.4)$$

be normalised entropy. Entropy is zero for a certain distribution and one for a uniform distribution. The implementation learns a minimum radius $r_{\min}$ and span s , initialised to 24 and 16 metres, and sets

$$r_k = \text{clip}(r_{\min}, 15, 35) + \text{clip}(s, 1, 25)(1 - p_k). \quad (3.5)$$

Low-confidence modes therefore inspect a wider endpoint neighbourhood. A learned sigmoid gate receives $(p_k, H(\mathbf{p}))$ and scales each mode’s target feature. The initial positive gate bias retains most of the baseline feature at the start of training.

3.5.5 Relative-geometry attention bias

Equation 2.12 is applied to each of the four scene-attention blocks. A 32-unit hidden layer predicts one bias per head from pairwise displacement, distance and heading difference. Zero initialisation makes the first forward pass equivalent to the baseline scores.

3.5.6 Kinematic motion stem

Masked velocity, acceleration and speed change are projected and added before temporal encoding. The hypothesis is that derivatives expose braking and acceleration directly. Derivatives also amplify observation noise and can duplicate information already represented by position and velocity, so this is treated as an empirical question.

3.5.7 Endpoint refinement

The coarse predicted endpoint is embedded and used to refine each trajectory and confidence query. An auxiliary loss with weight 0.2 supervises the coarse prediction. The design targets final displacement explicitly, but increases decoder coupling and may not improve average error.

3.5.8 Lane topology graph

For each lane token, up to eight nearby lanes within 40 metres are selected from geometry. A small edge network uses relative offset and heading to weight neighbour messages, followed by a residual update with LayerScale 0.01. This supplies sparse local connectivity without requiring new map labels.

3.6 Lab 3 attention study

The attention suite isolates the attention operator at a full 80-epoch horizon. It uses three RTX 2080 Ti GPUs, batch 8 per rank and global batch 24. The seed, data, model dimensions, non-attention blocks, losses and optimisation schedule are fixed. The suite replaces all 18 PyTorch multi-head attention constructors so that temporal, scene and target-context attention use the same operator in each variant.

The four variants are baseline MHA, QKNorm, Talking-Heads, and QKNorm with Talking-Heads. QKNorm applies Equation 2.10 independently per head. Talking-Heads uses identity-initialised pre-softmax and post-softmax mixing, ensuring that training starts close to standard MHA. The combined variant activates both mechanisms. Each run saves the lowest-minADE6 checkpoints; a separate validation artifact records the complete metric vector where available.

3.7 Final confirmatory three-run design

The final suite does not introduce another broad search. It combines only mechanisms selected by earlier evidence:

1. Official SHARP baseline. Pinned architecture, behaviour-preserving compatibility fixes and the article’s AV2 schedule.
2. QKNorm with uncertainty and geometry. All SHARP attention constructors use QKNorm. Uncertainty-aware target context uses Equation 3.5, while relative-geometry bias is added to the four scene-attention blocks.
3. Run 2 + residual temporal-agent Mamba. The small bidirectional residual module in Equation 3.2 is inserted between temporal-agent blocks 2 and 3.

The baseline receives no learned architectural addition. Behaviour-preserving compatibility changes replace an orthonormal matrix inverse by a transpose, construct index tensors on the correct CUDA device, use compatible mask dtypes, log the actual validation batch size and verify complete non-overlapping AdamW parameter groups. These changes remove runtime hazards without changing the intended model function.

The final suite uses a bounded NCCL collective test, a real one-GPU optimisation-step test (forward pass, backward pass and optimizer update), and a 256-step real-data four-GPU smoke test before full training. Every epoch writes `last.ckpt`. A failed variant retries from the newest loadable checkpoint up to three times. Completed variants are skipped on relaunch, and final validation uses one GPU with batch 32 to avoid distributed sample padding.

3.8 Reproducibility, logging and failure recovery

Long training jobs exposed practical issues that can otherwise invalidate an experiment: asynchronous illegal CUDA accesses, stale DDP ranks, NCCL transport faults, mask-dtype deprecations and ambiguous batch-size inference in metric logging. The recovery process follows four principles.

First, evidence is preserved before intervention. Logs, checkpoint inventories and process states are copied into timestamped recovery directories. Second, process matching is restricted to the exact experiment and result roots, preventing an unrelated user’s job from being stopped. Third, a checkpoint is loaded on CPU and its epoch and global step are inspected before resume. Fourth, a repair must pass an equivalence or real-data smoke test before multi-day training restarts.

Full terminal logs and checkpoints remain on the laboratory machines. Compact summaries, metrics, warning extracts, environment manifests and plots below the repository size threshold are merged to GitHub using token-based non-interactive authentication. No credential file is copied into a publication directory, and publication scripts scan staged files for token patterns before commit.

3.9 Research ethics and responsible reporting

The work uses public autonomous-driving datasets and does not collect personal data or interact with human participants. Dataset licences and source attribution remain applicable. The more immediate ethical issue is interpretation: small improvements in open-loop displacement error do not establish safer driving. The report therefore avoids claims about collision reduction or deployment readiness. Negative outcomes are retained, active metrics are not guessed, and all comparisons identify schedule, split and hardware differences. This restraint is necessary because a statistically small benchmark change can be overwhelmed by distribution shift, perception error or downstream planner behaviour in deployment.

4 RESULTS AND DISCUSSION

4.1 Reporting convention

All displacement and miss metrics are lower-is-better. “Best epoch” means the checkpoint with the lowest recorded validation minADE6. Values described as “final” are taken from the last validation epoch and are not silently treated as best-checkpoint values. The Lab 2 and Lab 3 controls differ in epoch count, global batch and hardware; comparisons across those studies are descriptive. Causal claims are limited to variants trained within the same controlled suite.

4.2 SHARP article and Lab 2 reproduction

Table 4.1 compares the SHARP article’s AV2 validation vector with the retained Lab 2 baseline. The local 60-epoch run does not reproduce the article’s 80-epoch schedule, and its global batch is 16 rather than 32. It is therefore evidence that the implementation trains successfully and approaches the published accuracy, not an exact reproduction claim.

Table 4.1: SHARP article AV2 validation result and retained Lab 2 reproduction.

Source	MR	b-minFDE6	minADE1	minADE6	minFDE1	minFDE6
SHARP article	0.140000	1.820000	1.570000	0.640000	3.850000	1.200000
Normal SHARP, Lab 2 (epoch 56)	0.150608	1.864411	1.599569	0.661463	3.962160	1.242391
Absolute gap (local – article)	+0.010608	+0.044411	+0.029569	+0.021463	+0.112160	+0.042391

The local minADE6 is 0.021463 higher, a relative gap of approximately 3.35%. Every listed metric is worse by a modest amount. Several factors plausibly contribute: 20 fewer epochs, a smaller global batch, DDP sample ordering, different GPU kernels and dependency versions, and possible preprocessing drift. The result should not be used to argue that four RTX 2080 Ti GPUs are intrinsically less accurate than one RTX 8000. Hardware affects the numerical execution and throughput; the schedule and data path affect the optimisation target more directly.

The article also reports AV2 test-set MR 0.14, b-minFDE6 1.83, minADE6 0.64 and minFDE6 1.19 [10]. These are test values and are not subtracted from local validation values.

4.3 Mamba placement in SHARP

Table 4.2 compares the longer Lab 2 Mamba additions with the local unmodified SHARP reproduction. The baseline uses a different 60-epoch/global-batch-16 setup, so the absolute comparison is not perfectly controlled. The direct scene-token versus temporal-agent comparison is stronger because both are 80-epoch, global-batch-32, four-GPU experiments.

Table 4.2: Longer Lab 2 SHARP and Mamba results on AV2 validation.

Model	Epoch evidence	MR	b-minFDE6	minADE1	minADE6	minFDE1	minFDE6
Normal SHARP	Best, 56/60	0.150608	1.864411	1.599569	0.661463	3.962160	1.242391
Scene-token Mamba	Best, 78/80	0.157353	1.926419	1.680364	0.680362	4.130003	1.290422
Residual temporal-agent Mamba	Final, 79/80	0.157000	1.920000	1.640000	0.676000	4.050000	1.290000

The residual temporal-agent run has a separately retained best minADE6 of 0.673736 at epoch 77. This is 0.006626 lower than scene-token Mamba, a 0.97% relative improvement, but remains 0.012273 above the local normal SHARP value. The ordering supports the placement hypothesis: chronological agent history is a more meaningful scan axis than pooled scene-token order. It does not support the stronger hypothesis that residual Mamba already improves SHARP accuracy.

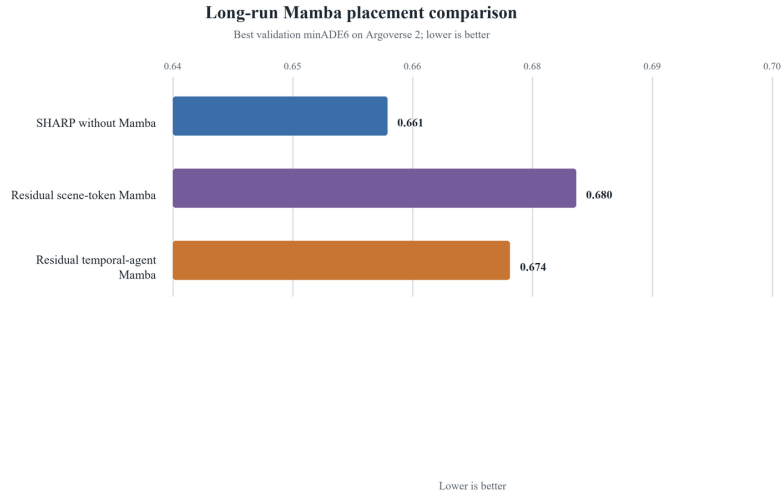


Figure 4.1: Best retained minADE6 for unmodified SHARP and the two longer Mamba additions. The baseline differs in schedule; scene-token and temporal-agent Mamba share the 80-epoch, global-batch-32 setting.

The 20-epoch complete replacement provides additional evidence. Replacing all four temporal attention blocks with Mamba gives minADE6 0.780190 against 0.752339 for the matched shortened baseline, a degradation of 0.027851 (3.70%). This is a controlled result against wholesale replacement in the tested configuration. Four attention blocks

can compare timesteps directly, while the replacement forces all temporal information through sequential state. With only ten observations per window, Mamba’s long-sequence scaling advantage is irrelevant and its inductive bias does not compensate for the removed attention.

4.4 Lab 2 20-epoch screening study

Table 4.3 presents all ten matched runs. The final column is variant minADE6 minus baseline; negative values are improvements.

Table 4.3: Controlled 20-epoch AV2 screening results.

Variant	Best ep.	MR	b-minFDE6	minADE6	minFDE6	Δ ADE
Baseline SHARP	19	0.192652	2.064007	0.752339	1.434819	+0.000000
Confidence-gated mem- ory	19	0.194413	2.067328	0.755349	1.443534	+0.003010
Cross-window consis- tency	18	0.202497	2.096022	0.776207	1.476766	+0.023868
Learned temporal pool- ing	19	0.201897	2.080306	0.757637	1.463850	+0.005298
Uncertainty-aware tar- get context	19	0.198575	2.074243	0.749961	1.447852	−0.002378
Relative-geometry attention bias	19	0.190852	2.051891	0.749679	1.432463	−0.002660
Kinematic motion stem	19	0.198175	2.068197	0.785164	1.448078	+0.032825
Endpoint refinement de- coder	19	0.196254	2.058200	0.755898	1.441044	+0.003559
Lane topology graph	19	0.200616	2.065051	0.754238	1.443290	+0.001899
Temporal Mamba re- placement	19	0.212822	2.141835	0.780190	1.514158	+0.027851

Only uncertainty-aware target context and relative geometry improve minADE6. Relative geometry is the strongest screen at 0.749679, 0.002660 lower than baseline (0.35%). It also improves MR from 0.192652 to 0.190852, b-minFDE6 from 2.064007 to 2.051891, minFDE1 from 4.672195 to 4.660576 and minFDE6 from 1.434819 to 1.432463. This consistency across endpoint and miss metrics makes it a more credible candidate than an isolated change in one column.

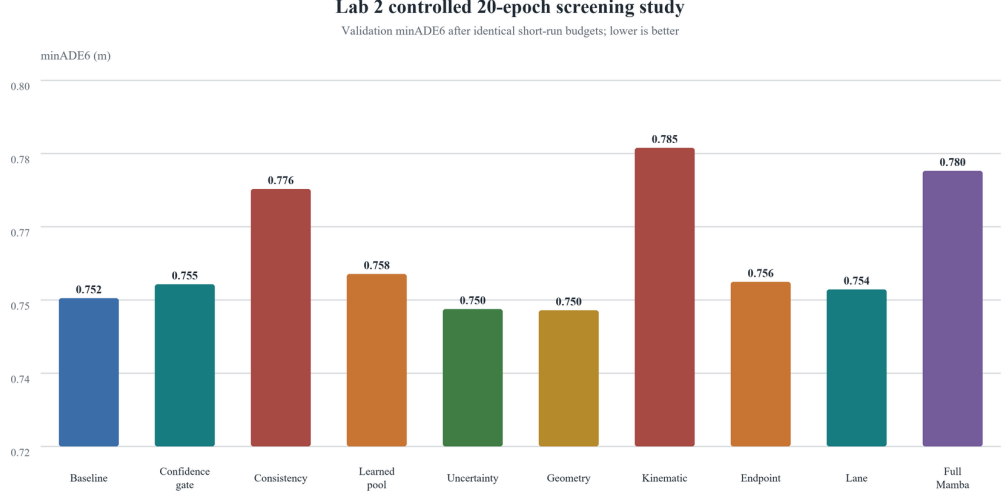


Figure 4.2: Best minADE6 in the Lab 2 20-epoch screen. The first bar is the matched shortened baseline; lower bars are better.

Uncertainty-aware target context improves minADE6 by 0.002378 (0.32%), while its MR and endpoint metrics are worse. A plausible interpretation is that adaptive context improves the average shape of at least one mode but does not yet improve endpoint coverage or confidence. It remains a candidate because its target-selection mechanism is complementary to geometry bias and its minADE6 change is in the desired direction.

The negative results are informative. The kinematic stem is worst at 0.785164, suggesting that derivative noise or redundant motion features are harmful under the short schedule. Cross-window consistency also degrades substantially. At early epochs, forcing agreement between uncertain consecutive predictions may preserve an error rather than stabilise a correct trajectory. Learned pooling and lane topology are close to baseline but do not justify extra complexity. Endpoint refinement worsens minADE6, yet improves minFDE1 from 4.672195 to 4.641530; it may benefit the single highest-priority trajectory without improving six-mode coverage.

The screen cannot establish final superiority because 13 of 20 epochs are warm-up. The selected variants may converge differently over 80 epochs, and selection on the same validation split introduces optimism. Its legitimate role is candidate reduction.

4.5 Lab 3 attention study

Table 4.4 gives the completed 80-epoch attention comparison. Best minADE6 is read from saved checkpoints. The remaining values use the retained validation vector; for the baseline MHA row, the final vector does not belong to the epoch-66 best-minADE6 checkpoint.

Table 4.4: Lab 3 attention results on AV2. All variants share the same 80-epoch, global-batch-24 controls.

Attention	Best ep.	Best ADE6	MR	b-minFDE6	minADE1	minFDE1	minFDE6	Δ ADE
Baseline MHA	66	0.673460	0.153231	1.927635	1.671288	4.063601	1.287932	+0.000000
QKNorm	65	0.669777	0.154471	1.899642	1.671904	4.119903	1.266240	-0.003683
Talking-Heads	71	0.674991	0.156510	1.927414	1.675181	4.130968	1.285642	+0.001531
QKNorm + Talking-Heads	71	0.674492	0.152064	1.931623	1.663308	4.090390	1.291059	+0.001032

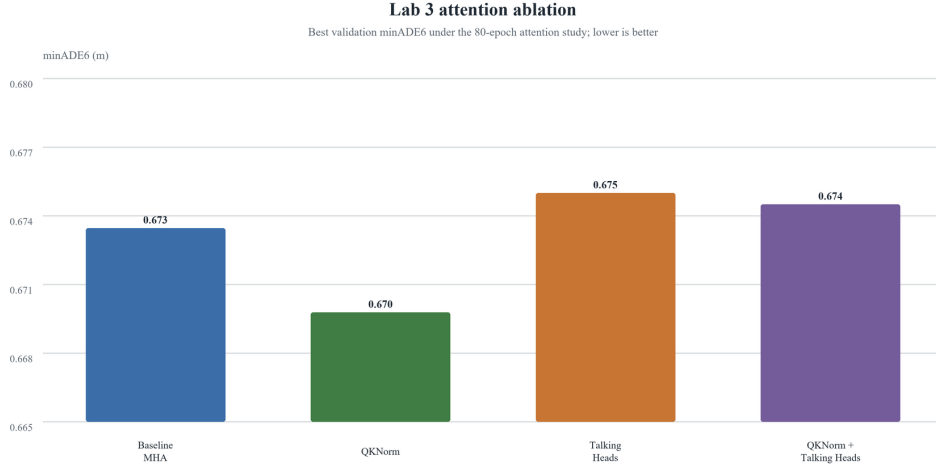


Figure 4.3: Best minADE6 for the Lab 3 attention operators. QKNorm is the only operator to improve the matched MHA control on this selection metric.

QKNorm improves minADE6 by 0.003683 (0.55%). It also lowers b-minFDE6 by 0.027993 and minFDE6 by 0.021692. MR is 0.001240 higher and minFDE1 is higher, so the improvement is not uniform. The pattern is compatible with QKNorm improving multimodal coverage and endpoint quality among six hypotheses without improving the single-mode forecast.

Talking-Heads is 0.001531 worse than MHA on minADE6 and does not lead any retained metric. Combining it with QKNorm is 0.001032 worse than MHA, despite giving the lowest MR (0.152064) and minADE1 (1.663308). The combined operator may distribute modes in a way that avoids complete misses while slightly increasing oracle average dis-

placement. This is a useful example of why one metric cannot summarise every desirable property.

The strongest conclusion from Lab 3 is narrow: under this implementation and seed, QKNorm is the appropriate attention candidate for the final minADE6-focused comparison. There is no evidence that arbitrary head mixing should also be retained.

4.6 Cross-dataset Lab 2 evidence

The normal Lab 2 SHARP checkpoint was also evaluated on Argoverse 1 and nuScenes. Table 4.5 records these values beside the article. The comparisons are descriptive because the nuScenes split and number of modes differ.

Table 4.5: Secondary Lab 2 validation evidence and SHARP article values.

Dataset	Source	Evaluation	MR	minADE _K	minFDE1	minFDE _K
Argoverse 1	Article	Validation, $K = 6$	0.070000	0.580000	–	0.900000
Argoverse 1	Lab 2 SHARP	Validation, $K = 6$	0.088670	0.635403	2.765118	1.027457
nuScenes	Article	Challenge test, $K = 5$	0.280000	1.130000	6.020000	–
nuScenes	Lab 2 SHARP	Validation, $K = 6$	0.302300	1.150025	6.344474	1.965479

The local Argoverse 1 minADE6 is 0.055403 higher than the article and minFDE6 is 0.127457 higher. The local nuScenes minADE is close in absolute terms, but comparing $K = 6$ validation to $K = 5$ challenge test is not controlled. These results reinforce the need to describe preprocessing, split and mode count whenever a benchmark number is reused.

4.7 Training-time context

Table 4.6: Approximate completed-run wall-clock time. Article training time was not reported.

Run	Epochs	Global batch	Approx. elapsed time
Normal SHARP, Lab 2	60	16	109 h
Baseline MHA, Lab 3	80	24	99 h 39 min
Scene-token Mamba, Lab 2	80	32	129 h 17 min
Temporal-agent Mamba, Lab 2	80	32	160 h 20 min

The number of GPUs does not determine wall-clock time alone. DDP communication, validation cadence, synchronized BatchNorm, data loading, desktop display load and the additional Mamba computation all contribute. Temporal-agent Mamba is the slowest retained run despite using four GPUs. Its bidirectional selective scans and compaction work add computation to the original four attention blocks rather than replacing them.

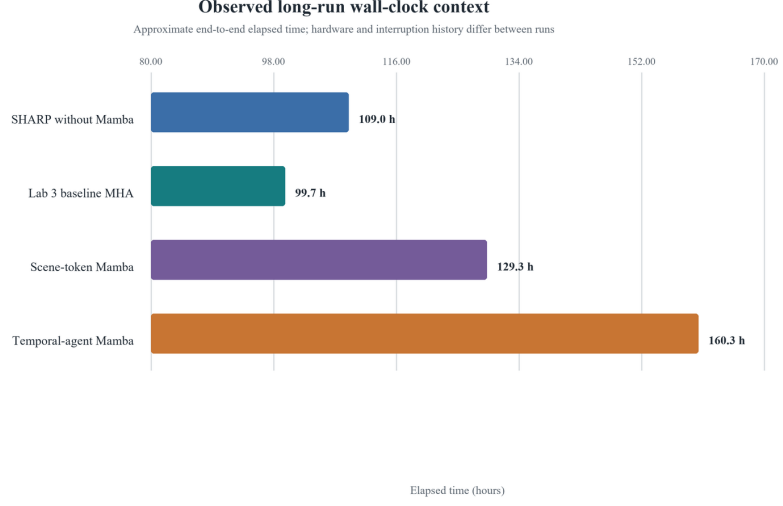


Figure 4.4: Approximate wall-clock hours for completed Lab 2 and Lab 3 reference runs. Differences include model, epoch, batch and system-load effects.

4.8 Final confirmatory study: reserved results

The final suite tests whether the individually supported changes remain beneficial under one 80-epoch, global-batch-32 protocol. At the dissertation snapshot used to create this draft, the suite has passed its configuration and distributed preflights but has not produced three validated final vectors. Table 4.7 and Figure 4.5 intentionally reserve those outputs. A value must be inserted only from the suite’s post-training, single-GPU validation artifact.

Table 4.7: Final three-run confirmatory comparison. Pending cells must be populated from validated artifacts.

Configuration	MR	b-minFDE6	minADE1	minADE6	minFDE1	minFDE6
Official SHARP baseline			Pending final validated result			
QKNorm + uncertainty + geometry			Pending final validated result			
Run 2 + residual temporal-agent Mamba			Pending final validated result			

The expected analysis is pre-specified. Run 2 is compared with run 1 to test the combined effect of QKNorm, uncertainty-aware context and geometry. Run 3 is compared with run 2 to isolate the incremental residual Mamba effect. It would be invalid to compare run 3 only with the article because any difference would combine four modifications and an execution-environment difference. The final report should include absolute deltas, relative percentage deltas and all six metrics, not only the best-looking column.

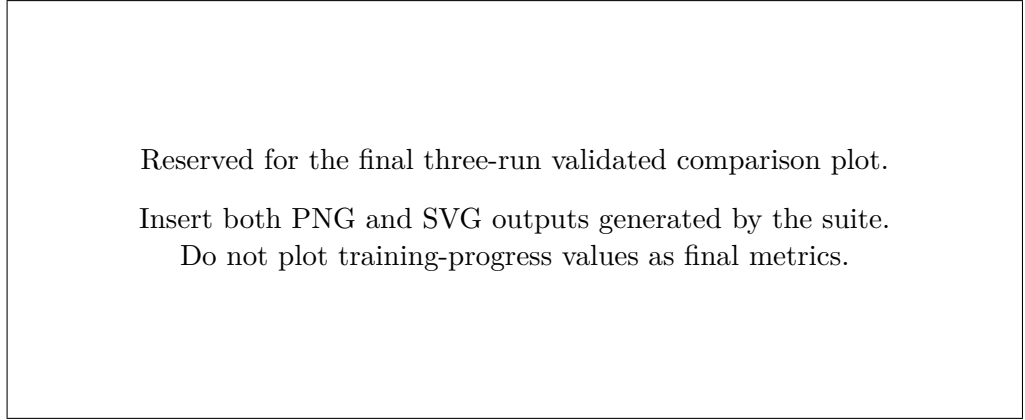


Figure 4.5: Reserved confirmatory-suite comparison.

4.9 Answers to the research questions

RQ1: reproduction. The retained Lab 2 baseline approaches but does not match the article. Its minADE6 is 0.661463 versus 0.640000, with identifiable schedule and execution differences. The final article-aligned baseline is necessary before a stronger reproduction claim can be made.

RQ2: broad modifications. Two of nine modifications improve the matched 20-epoch baseline on minADE6: relative-geometry bias and uncertainty-aware target context. Relative geometry gives the most consistent multi-metric improvement. Most extra modules are neutral or harmful under the short schedule.

RQ3: attention. QKNorm gives the best minADE6 and endpoint metrics among the four matched Lab 3 attention variants. Talking-Heads does not improve minADE6, either alone or with QKNorm, although the combined operator gives the lowest MR.

RQ4: Mamba. The residual temporal-agent addition is better than scene-token Mamba, which supports chronological placement, but neither longer Mamba addition beats the local unmodified SHARP reproduction. Complete replacement is clearly worse in the matched 20-epoch screen. The evidence favours residual use as a small auxiliary pathway, not replacement of every temporal attention block.

4.10 Threats to validity

The experiments use one principal seed. The small improvements observed for QKNorm, uncertainty and geometry may lie within seed-to-seed variance; three-seed replication

would be required for a statistical claim. The 20-epoch schedule is dominated by warm-up and may rank models differently from 80 epochs. Candidate selection and evaluation use the same validation split, which introduces selection bias. Some long-run tables combine a best minADE6 checkpoint with other metrics from a final validation vector; those rows are labelled but are less complete than evaluating every best checkpoint identically.

Hardware and software differ from the article. Distributed reduction, sample order and kernels can produce diverging trajectories despite the same seed and global batch. The local normal baseline also differs in epoch count and batch, so it cannot serve as a fully controlled comparator for the longer Mamba additions. Finally, open-loop oracle metrics do not measure calibration, temporal stability, collisions or planner utility. These limitations constrain the conclusions to the observed benchmark and configurations.

5 CONCLUSIONS AND FUTURE WORK

5.1 Summary of the investigation

This dissertation investigated whether SHARP can be improved through targeted changes to attention, target context and temporal sequence modelling without removing its defining short-window streaming mechanism. The work began with an article and source audit, because “original SHARP” was not a single unambiguous configuration: the article specifies an 80-epoch AV2 schedule with peak learning rate 10^{-4} , while the audited repository configuration had drifted to 60 epochs and 10^{-3} . That distinction shaped the experimental method. Local results were classified by the setup actually executed, and the final confirmatory baseline was pinned to the released architecture while restoring the article schedule.

The broad Lab 2 screen showed that added complexity is not a reliable proxy for accuracy. Only uncertainty-aware target context and relative-geometry bias improved minADE6 over the matched 20-epoch control. Relative geometry was the strongest screen and improved MR and endpoint metrics as well. In contrast, explicit kinematic derivatives, cross-window consistency and complete temporal Mamba replacement caused substantial degradation. Learned temporal pooling, lane topology and endpoint refinement were close enough to the baseline to remain conceptually interesting, but did not justify inclusion in the final combination.

The Lab 3 attention study provided a clearer full-length result. QKNorm reduced best minADE6 from 0.673460 to 0.669777 and improved b-minFDE6 and minFDE6. Talking-Heads attention did not improve minADE6, either alone or with QKNorm. The combined operator did produce the best MR, showing that attention variants can alter mode coverage and average displacement differently.

The Mamba experiments establish a placement-sensitive negative result. Residual bidirectional Mamba inside the chronological agent encoder is better than scanning pooled scene tokens, but the best retained value, 0.673736, does not beat the 0.661463 local unmodified SHARP reproduction. Replacing all four temporal attention blocks is worse

still. The tested SHARP window contains only ten observations, so Mamba’s long-sequence efficiency is not decisive; direct temporal attention remains useful. Mamba is therefore best treated as a small residual hypothesis to test after empirically supported attention and geometry changes, rather than as a wholesale substitute.

5.2 Contributions

The study contributes an evidence-backed modification map for SHARP. It identifies which changes act on temporal history, scene interaction, target context or streaming memory, and records whether each mechanism improved the matched control. This prevents conclusions such as “Mamba improves SHARP” from obscuring the more accurate statement that one Mamba placement was less harmful than another but neither surpassed the retained baseline.

It also contributes a reproducible experimental workflow. Long runs use isolated directories, exact code copies, environment manifests, per-epoch last checkpoints, CPU checkpoint validation, bounded real-data preflight tests and retry logic. Compact result artifacts are versioned separately from full logs and checkpoints. This operational work is part of the scientific contribution because an interrupted or ambiguously resumed run cannot support a reliable ablation.

Finally, the dissertation connects the completed screens to a pre-specified confirmatory study. The three planned rows isolate the article-aligned baseline, the combined evidence-supported QKNorm–uncertainty–geometry model, and the incremental residual Mamba effect. Reserving the result cells until validation prevents an active progress value from becoming a thesis conclusion.

5.3 Practical conclusions

Four practical conclusions follow from the evidence.

First, preserve the streaming innovation when testing local operators. None of the reported ablations removes short-window processing, identity-aware context, target context or trajectory relay. A model that improves a snapshot metric by discarding those mechanisms would answer a different question.

Second, add motion-specific inductive bias where the relation is explicit. Relative displacement and heading are directly relevant to traffic interaction, and the geometry bias improves the short screen with little disruption to the baseline. Its zero-initialised output also makes the intervention conservative.

Third, stabilise attention before increasing interaction capacity. QKNorm improves the selected metric, while Talking-Heads mixing does not. This suggests that controlling query-key scale is more useful in this compact network than allowing heads to communicate through additional learned projections.

Fourth, use Mamba only on a semantically ordered axis and retain a residual path. Chronological agent observations satisfy that requirement; pooled scene tokens do not. Even then, the sequence is short and attention is strong, so Mamba should enter through a small LayerScale rather than replace the complete encoder.

5.4 Limitations

The principal limitation is statistical. Most controlled runs use one seed, and the best positive deltas are small. The work identifies promising candidates but cannot estimate confidence intervals. The shortened screen also preserves a 13-epoch warm-up within 20 epochs, which is useful for matched selection but not representative of full convergence.

The article execution environment was not reproduced exactly. Four RTX 2080 Ti GPUs replace one RTX 8000, and dependency versions and CUDA kernels differ. Although global batch and learning-rate controls are explicit, bitwise or exact-metric reproduction cannot be guaranteed. Secondary Argoverse 1 and nuScenes comparisons also involve split or mode-count differences.

The evaluation is open loop. minADE, minFDE and MR do not measure whether a planner selects safer actions, whether predictions remain calibrated under distribution shift, or whether trajectories fluctuate between consecutive streaming windows. No closed-loop simulation or on-vehicle experiment is claimed.

5.5 Future work

The immediate task is to complete and validate the three final 80-epoch runs. Each best checkpoint should be evaluated with the same single-GPU batch-32 script, and Table 4.7 should then be filled from machine-readable JSON rather than terminal progress text. The comparison should report all six metrics and confidence intervals across at least three seeds if compute permits.

If run 2 improves the article-aligned baseline, a factorial follow-up should separate QKNorm, uncertainty and geometry at 80 epochs. The current screen shows individual effects for uncertainty and geometry and an independent Lab 3 effect for QKNorm, but the combination may be additive, redundant or antagonistic. A 2^3 design would estimate interactions directly.

If residual Mamba fails to improve run 2, further Mamba work should not increase depth indiscriminately. More informative tests would vary residual scale, use a single forward scan to respect online causality, or share state across SHARP windows rather than rescan ten observations. The last option is architecturally more aligned with SHARP’s streaming objective, but it requires careful identity lifecycle handling and detach rules to prevent unbounded computation graphs.

Evaluation should also move beyond oracle displacement. Calibration metrics, negative log-likelihood, diversity, collision rate and prediction consistency across consecutive windows would test properties hidden by minADE6. Closed-loop CARLA or log-replay planning experiments could then measure whether metric changes affect braking, comfort and progress. Such a study would be a new systems contribution and should not be inferred from the present benchmark alone.

5.6 Final statement

The central lesson is methodological: a promising modern component improves a mature forecasting model only when its inductive bias matches the representation and when the comparison is controlled. In this study, explicit geometry and query–key normalisation provide the clearest positive evidence. Mamba is most coherent inside the ordered agent-history encoder, but its tested forms do not yet exceed unmodified SHARP. The resulting final design is therefore conservative: preserve SHARP’s streaming structure, retain direct

attention, add only evidence-supported context and geometry, and treat residual Mamba as the final isolated question rather than a predetermined success.

BIBLIOGRAPHY

- [1] Alahi, A. , Goel, K. , Ramanathan, V. , Robicquet, A. , Fei-Fei, L. , and Savarese, S. . Social lstm: Human trajectory prediction in crowded spaces. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pages 961–971, 2016. URL https://openaccess.thecvf.com/content_cvpr_2016/html/Alahi_Social_LSTM_Human_CVPR_2016_paper.html.
- [2] Caesar, H. , Bankiti, V. , Lang, A. H. , Vora, S. , Liong, V. E. , Xu, Q. , Krishnan, A. , Pan, Y. , Baldan, G. , and Beijbom, O. . nuscenes: A multimodal dataset for autonomous driving. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 11621–11631, 2020. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Caesar_nuScenes_A_Multimodal_Dataset_for_Autonomous_Driving_CVPR_2020_paper.html.
- [3] Chang, M.-F. , Lambert, J. , Sangkloy, P. , Singh, J. , Bak, S. , Hartnett, A. , Wang, D. , Carr, P. , Lucey, S. , Ramanan, D. , and Hays, J. . Argoverse: 3d tracking and forecasting with rich maps. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 8748–8757, 2019. URL https://openaccess.thecvf.com/content_CVPR_2019/html/Chang_Argoverse_3D_Tracking_and_Forecasting_With_Rich_Maps_CVPR_2019_paper.html.
- [4] Gao, J. , Sun, C. , Zhao, H. , Shen, Y. , Anguelov, D. , Li, C. , and Schmid, C. . Vectornet: Encoding hd maps and agent dynamics from vectorized representation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 11525–11533, 2020. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Gao_VectorNet_Encoding_HD_Maps_and_Agent_Dynamics_From_Vectorized_Representation_CVPR_2020_paper.html.
- [5] Gu, A. and Dao, T. . Mamba: Linear-time sequence modeling with selective state spaces. In First Conference on Language Modeling, 2024. URL <https://openreview.net/forum?id=tEYskw1VY2>.
- [6] Gupta, A. , Johnson, J. , Fei-Fei, L. , Savarese, S. , and Alahi, A. . Social gan: Socially acceptable trajectories with generative adversarial networks. In Proceedings

- of the IEEE Conference on Computer Vision and Pattern Recognition, pages 2255–2264, 2018. URL https://openaccess.thecvf.com/content_cvpr_2018/html/Gupta_Social_GAN_Socially_CVPR_2018_paper.html.
- [7] Henry, A. , Dachapally, P. R. , Pawar, S. S. , and Chen, Y. . Query-key normalization for transformers. In Findings of the Association for Computational Linguistics: EMNLP 2020, pages 4246–4253, 2020. doi: 10.18653/v1/2020.findings-emnlp.379. URL <https://aclanthology.org/2020.findings-emnlp.379/>.
- [8] Liang, M. , Yang, B. , Hu, R. , Chen, Y. , Liao, R. , Feng, S. , and Urtasun, R. . Learning lane graph representations for motion forecasting. In Computer Vision – ECCV 2020, pages 541–556, 2020. doi: 10.1007/978-3-030-58536-5_32. URL <https://arxiv.org/abs/2007.13732>.
- [9] Loshchilov, I. and Hutter, F. . Decoupled weight decay regularization. In International Conference on Learning Representations, 2019. URL <https://openreview.net/forum?id=Bkg6RiCqY7>.
- [10] Prutsch, A. , Fruhwirth-Reisinger, C. , Schinagl, D. , and Possegger, H. . SHARP: Short-window streaming for accurate and robust prediction in motion forecasting. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 32103–32112, June 2026. URL https://openaccess.thecvf.com/content/CVPR2026/html/Prutsch_SHARP_Short-Window_Streaming_for_Accurate_and_Robust_Prediction_in_Motion_Forecasting_CVPR_2026_paper.html.
- [11] Prutsch, A. , Schinagl, D. , and Possegger, H. . Streaming real-time trajectory prediction using endpoint-aware modeling. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pages 3005–3014, 2026. URL https://openaccess.thecvf.com/content/WACV2026/html/Prutsch_Streaming_Real-Time_Trajectory_Prediction_Using_Endpoint-Aware_Modeling_WACV_2026_paper.html.
- [12] Salehi, S. M. . Thesis experiment repository: Sharp reproduction, attention and mamba ablations. GitHub repository, 2026. URL <https://github.com/madvidd/Thesis>.

- [13] Salzmann, T. , Ivanovic, B. , Chakravarty, P. , and Pavone, M. . Trajectron++: Dynamically-feasible trajectory forecasting with heterogeneous data. In Computer Vision – ECCV 2020, pages 683–700, 2020. URL https://www.ecva.net/papers/eccv_2020/papers_ECCV/html/3094_ECCV_2020_paper.php.
- [14] Shaw, P. , Uszkoreit, J. , and Vaswani, A. . Self-attention with relative position representations. In Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2, pages 464–468, 2018. doi: 10.18653/v1/N18-2074. URL <https://aclanthology.org/N18-2074/>.
- [15] Shazeer, N. , Lan, Z. , Cheng, Y. , Ding, N. , and Hou, L. . Talking-heads attention. arXiv preprint arXiv:2003.02436, 2020. URL <https://arxiv.org/abs/2003.02436>.
- [16] Song, N. , Zhang, B. , Zhu, X. , and Zhang, L. . Motion forecasting in continuous driving. In Advances in Neural Information Processing Systems, volume 37, 2024. doi: 10.52202/079017-2484. URL https://papers.nips.cc/paper_files/paper/2024/hash/8ec078518dcce6be1324cfd3de11ed24-Abstract-Conference.html.
- [17] Vaswani, A. , Shazeer, N. , Parmar, N. , Uszkoreit, J. , Jones, L. , Gomez, A. N. , Kaiser, L. , and Polosukhin, I. . Attention is all you need. In Advances in Neural Information Processing Systems, volume 30, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.
- [18] Wilson, B. , Qi, W. , Agarwal, T. , Lambert, J. , Singh, J. , Khandelwal, S. , Pan, B. , Kumar, R. , Hartnett, A. , Pontes, J. K. , Ramanan, D. , Carr, P. , and Hays, J. . Argoverse 2: Next generation datasets for self-driving perception and forecasting. In Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks, 2021. URL <https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/4734ba6f3de83d861c3176a6273cac6d-Abstract-round2.html>.
- [19] Zhang, B. , Song, N. , and Zhang, L. . DeMo: Decoupling motion forecasting into directional intentions and dynamic states. In Advances in Neural Information Processing Systems, volume 37, 2024. doi: 10.52202/079017-3383. URL https://papers.nips.cc/paper_files/paper/2024/hash/c0ff9e52e94ae331bc0f2d28be06a9ca-Abstract-Conference.html.

- [20] Zhou, Z. , Ye, L. , Wang, J. , Wu, K. , and Lu, K. . HiVT: Hierarchical vector transformer for multi-agent motion prediction. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 8823–8833, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Zhou_HiVT_Hierarchical_Vector_Transformer_for_Multi-Agent_Motion_Prediction_CVPR_2022_paper.html.
- [21] Zhou, Z. , Wang, J. , Li, Y.-H. , and Huang, Y.-K. . Query-centric trajectory prediction. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 17863–17873, 2023. URL https://openaccess.thecvf.com/content/CVPR2023/html/Zhou_Query-Centric_Trajectory_Prediction_CVPR_2023_paper.html.

6 REPRODUCIBILITY AND SUPPORTING EVIDENCE

6.1 Template and source layout

The dissertation is stored under `Dissertation/` in the Thesis repository. `Dissertation_Template.tex` remains the main Overleaf document. The original template’s parameters/ and style/ files are copied without formatting changes. Dissertation text is contained in `chapters/`, references in `references/thesisbib.bib`, and generated figures in `images/`.

The project evidence used to prepare the dissertation is retained under the following repository locations:

- `Results/Main/SHARP_AV2_Main_Results.md`: consolidated article and local metric vectors;
- `Results/Main/All_Run_Registry.md`: run status and evidence links;
- `Results/Info.md`: architectural modification glossary;
- `Lab2/SHARP_20_Epoch_10_Test_Suite/`: controlled screening code, manifests and completed results;
- `Lab2/SHARP_Final_3_Run_Suite/`: article-aligned final protocol, audited runtime modules and active progress;
- `Lab3/Main_Results/Attention_Experiments/`: completed attention artifacts; and
- `Lab3/Codes/`: attention runtime preparation, recovery and publication scripts.

Operational snapshots were consulted to establish what ran, but restarts and publication attempts were not counted as separate experiments.

6.2 Full Lab 2 screening metric vector

Table 6.1: Full retained metric vector for the 20-epoch screen.

Variant	MR	b-FDE6	ADE1	ADE6	FDE1	FDE6	Δ ADE6
Baseline	.192652	2.064007	1.886442	.752339	4.672195	1.434819	.000000

Variant	MR	b-FDE6	ADE1	ADE6	FDE1	FDE6	Δ ADE6
Confidence gate	.194413	2.067328	1.876097	.755349	4.694828	1.443534	+.003010
Window consistency	.202497	2.096022	1.872110	.776207	4.785388	1.476766	+.023868
Learned pooling	.201897	2.080306	1.891240	.757637	4.739688	1.463850	+.005298
Uncertainty context	.198575	2.074243	1.863236	.749961	4.676969	1.447852	−.002378
Geometry bias	.190852	2.051891	1.875754	.749679	4.660576	1.432463	−.002660
Kinematic stem	.198175	2.068197	1.942671	.785164	4.701560	1.448078	+.032825
Endpoint refinement	.196254	2.058200	1.867779	.755898	4.641530	1.441044	+.003559
Lane topology	.200616	2.065051	1.903412	.754238	4.696756	1.443290	+.001899
Temporal Mamba re- placement	.212822	2.141835	1.993448	.780190	4.973433	1.514158	+.027851

6.3 Mamba implementation parameters

Table 6.2: Residual temporal-agent Mamba implementation.

Property	Configuration
Insertion	Between temporal attention blocks 2 and 3
Input feature dimension	128
Direction	Separate forward and backward modules
d_{state}	8
Local convolution width	3
Expansion ratio	1
Normalisation	Pre-Mamba LayerNorm
Fusion	Per-channel sigmoid gate
Dropout	0.1
LayerScale initial value	0.01
Agent chunk size	128
Padding policy	Compact valid observations, restore afterwards
CUDA path	Fused selective scan and cuDNN Conv1d

6.4 Completed Lab 3 checkpoint evidence

Table 6.3: Best-minADE6 checkpoint evidence for the attention suite.

Attention	Best epoch	Best minADE6	Delta vs MHA
MHA	66	0.673460	+0.000000
QKNorm	65	0.669777	−0.003683
Talking-Heads	71	0.674991	+0.001531
QKNorm + Talking-Heads	71	0.674492	+0.001032

6.5 Final-suite validation checklist

The following checklist prevents active training values from being promoted into the final result table:

1. confirm that all three variant directories contain a completion marker and a loadable last.ckpt;
2. select the checkpoint with minimum validation minADE6 according to the same rule for every run;
3. run the final evaluation script on one GPU with batch 32, avoiding DDP validation padding;
4. obtain MR, b-minFDE6, minADE1, minADE6, minFDE1 and minFDE6 from the generated machine-readable JSON;
5. verify that the Markdown/CSV table and plotted values match the JSON to six decimal places;
6. insert the three vectors into Table 4.7;
7. replace the reserved Figure 4.5 with the suite’s PNG while retaining the SVG alongside it;
8. update the abstract, conclusion, word count and page count; and
9. compile twice after BibTeX and inspect every page before submission.

6.6 Reproduction caveats

The word “reproduction” is used with explicit qualification in this dissertation. An exact floating-point match is not guaranteed when GPU architecture, CUDA libraries, distributed execution and data-loader order differ. The final baseline is article-aligned in architecture and optimisation controls, while the paper’s execution used one RTX 8000. The retained earlier Lab 2 baseline is a 60-epoch local reproduction with global batch 16. The Lab 3 attention baseline is an 80-epoch internal control with global batch 24. These are distinct evidence categories.

6.7 Reference and link validation

Bibliographic entries prioritise primary conference pages, official proceedings, DOI records and author repositories. The most important external records are the CVPR page for SHARP [10], the NeurIPS pages for DeMo and RealMotion [19, 16], the WACV page for

SEAM [11], OpenReview for Mamba and AdamW [5, 9], ACL Anthology for QKNorm and relative-position attention [7, 14], and official dataset proceedings for AV2, Argoverse 1 and nuScenes [18, 3, 2]. URLs are included in BibTeX so entries are clickable in the generated PDF.

6.8 Suggested final artifact set

For dissertation submission, retain the compiled PDF, all LaTeX sources, BibTeX database, PNG/SVG figures, exact result JSON/CSV files, environment manifests and the final code commit. Full training logs and checkpoints are useful archival evidence but are not required inside the Git repository and should remain in their laboratory archive because of size.